

Performance Metrics

CSC3034 Computational Intelligence

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Introduction

Gold Standard

Percent Correct

Errors

Average Sum-Squared Error

Absolute Error

Normalised Error

Mann-Whitney U Test

Matrices

Contingency Matrices

Confusion Matrices

Receiver Operating Characteristic (ROC) Curves

Chi-Square Test

Under this topic, we will look at different metrics to evaluate the performance of a computational intelligence system.

- ▶ percent correct
- ▶ errors
- ▶ evolutionary algorithm effectiveness metrics
- ▶ Mann-Whitney U test
- ▶ contingency/confusion matrices
- ▶ receiver operating characteristic (ROC) curves
- ▶ chi-square test

- ▶ Gold standard is the 'correct output' or 'right answer'.
- ▶ For example in supervised neural network trainings, the data is provided in the form of input patterns and expected output patterns. The data of expected output patterns is the gold standard.
- ▶ Gold standard is used to compare with the output of a system to evaluate the performance of the system.

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Percent correct

Percent correct is the percentage of all answers that were judged to be correct according to some gold standard.

The limitation of percent correct can be seen through the following example:

Suppose we have a group of 100 stocks. A computational intelligence tool is used to predict whether the stock will outperform or perform less well than the Dow Jones average.

Last month, 90 percent of the stocks that outperform were predicted accurately, and 60 percent of the stocks that perform less well were predicted correctly.

This month, 85 percent of the stocks that outperform were predicted accurately, and 55 percent of the stocks that perform less well were predicted correctly.

The percent correct decreased, however, the overall ability to predict could have improved.

How is this possible?

Suppose half of the stocks last month outperformed the Dow Jones average,

$$\text{overall performance} = \frac{0.9 \times 50 + 0.6 \times 50}{100} = 0.75$$

and 70 percent of the stocks this month outperformed the Dow Jones average,

$$\text{overall performance} = \frac{0.85 \times 70 + 0.55 \times 30}{100} = 0.76$$

The overall predictive accuracy increased, even though the predictive accuracy on the individual metrics decreased.

What the example is showing is that,

- ▶ the percent correct can be misleading if it is the only method of evaluating performance, and
- ▶ we might want to consider more than one performance metric in evaluating a system.

Note: In neural network implementations, we often establish ranges within which answers are considered correct.

For example, if the output can vary from 0 to 1, we might want to define such that any output in the range of

0.8 to 1.0 is considered 1
0.0 to 0.2 is considered 0

In more extreme case, we could have

0.5 to 1.0 considered as 1
0.0 to 0.5 considered as 0

This manipulation of percent correct metric introduces subjectivity, but improve its usefulness in practical sense.

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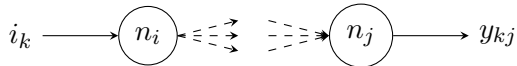
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Average sum-squared error

Average sum-squared error is often used in evaluated the performance of a neural networks to determine if the training should be terminated.

Consider b_{kj} as the expected output value of neuron j for pattern k



$$\text{total error, } E_t = 0.5 \sum_{k=1}^m \sum_{j=1}^q (b_{kj} - y_{kj})^2$$

$$\text{average sum-squared error, } e = \frac{E_t}{m}$$

where m is the total number of patterns

q is the total number of output neurons

Average sum-squared error

$$\text{total error, } E_t = 0.5 \sum_{k=1}^m \sum_{j=1}^q (b_{kj} - y_{kj})^2$$

Note that the original definition by Rumelhart and McClelland (1986) includes the multiplier 0.5, which allow 'neat' differentiation that simplifies the math elsewhere.

Many implementations ignore this factor to reduce calculation time.

Therefore it is important to be aware of how the error term is calculated when you are using average sum-squared error to compare networks.

Average sum-squared error

The error term is summed over all output neurons, and averaged over the number of patterns.

Therefore comparing networks with different number of output neurons using average sum-squared error might be misleading.

One might want to compute the error per output neuron.

$$e \text{ per output neuron} = \frac{\text{average sum-squared error, } e}{\text{number of output neurons, } q}$$

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Absolute error is more intuitive to human compare to the average sum-squared error.

Mean absolute error

$$E_{ma} = \frac{1}{mq} \sum_{k=1}^m \sum_{j=1}^q |b_{kj} - y_{kj}|$$

where m is the total number of patterns

q is the total number of output neurons

Maximum absolute error

$$E_{\max} = \max_{k,j} |b_{kj} - y_{kj}|$$

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Normalised error takes into consideration the variance of the output patterns. By doing so, it becomes independent of the variance.

$$\text{total error, } E_t = 0.5 \sum_{k=1}^m \sum_{j=1}^q (b_{kj} - y_{kj})^2$$

$$\text{mean error, } E_{\text{mean}} = 0.5 \sum_{k=1}^m \sum_{j=1}^q (b_{kj} - \mu_j)^2$$

$$\text{where } \mu_j = \frac{1}{m} \sum_{k=1}^m b_{kj}$$

$$\text{normalised error, } E_n = \frac{E_t}{E_{\text{mean}}}$$

One way to look at the normalised error is that it reflects the proportion of the output variance that is due to error rather than the architecture of the network itself.

Overall it is believed that this error measure may be the most useful one for back-propagation neural network implementations.

The issue with normalised error happens when the expected output of a neuron does not change for all patterns. The expected output always equals to the mean value.

$$\text{mean error, } E_{\text{mean}} = 0.5 \sum_{k=1}^m \sum_{j=1}^q (b_{kj} - \mu_j)^2$$

$b_{kj} - \mu_j$ becomes 0 for that neuron, and therefore decreases E_{mean} .

This may cause the normalised error E_n to be 'artificially' larger than it should be.

To overcome this situation, it is suggested to remove the output neurons whose expected output does not change value.

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- ▶ Also known as Mann-Whitney-Wilcoxon test or the Wilcoxon rank-sum test.
- ▶ It is used to analyse the performance of evolutionary algorithms.
- ▶ The test evaluates whether the medians of two samples of data are the same using hypothesis testing.
- ▶ The null hypothesis is that the medians of the two samples are equal.
- ▶ The test will inform us to reject or not to reject the null hypothesis.

The two steps in a Mann-Whitney U test are

1. to identify the U value, and
2. to determine if the null hypothesis should be rejected.

Calculation of the U value:

Method 1 quick and direct

Method 2 formula

Consider

- ▶ we are running a minimisation problem with an optimum of 0.0, (lower fitness is better) and
- ▶ we are using an evolutionary algorithm with two configurations A and B, where B is our “baseline” configuration.

5 runs are made with configuration A and obtain the best fitness values of

0.079 0.062 0.073 0.047 0.085

and 4 runs are made with configuration B and obtain the best fitness values of

0.102 0.069 0.055 0.049

n_1 and n_2 are number of runs made with configurations A and B respectively.

$$\therefore n_1 = 5, n_2 = 4$$

Next, we arrange the measurements in the order of fitness with better fitness on the left.

0.047	0.049	0.055	0.062	0.069	0.073	0.079	0.085	0.102
A	B	B	A	B	A	A	A	B

Method 1: quick and direct

1. count the number of A entries that are better than each of the B entries.

	1 st B	2 nd B	3 rd B	4 th B	
# of A better than	1	1	2	5	$U_1 = \text{Total} = 9$

2. count the number of B entries that are better than each of the A entries.

	1 st A	2 nd A	3 rd A	4 th A	5 th A	
# of B better than	0	2	3	3	3	$U_2 = \text{Total} = 11$

3. The smaller of U_1 and U_2 is the U value, i.e. 9.

* Note that the two values add up to $n_1 n_2$.

0.047	0.049	0.055	0.062	0.069	0.073	0.079	0.085	0.102
A	B	B	A	B	A	A	A	B

Method 2: formula

1. label the rank from 1 to 9 going from left to right.
2. calculate R_1 , the sum of ranks for A's, and U_1 .

$$R_1 = 1 + 4 + 6 + 7 + 8 = 26$$

$$U_1 = n_1 n_2 + \frac{n_1(n_1 + 1)}{2} - R_1 = 5(4) + \frac{5(5 + 1)}{2} - 26 = 9$$

3. calculate R_2 , the sum of ranks for B's, and U_2 .

$$R_2 = 2 + 3 + 5 + 9 = 19$$

$$U_2 = n_1 n_2 + \frac{n_2(n_2 + 1)}{2} - R_2 = 5(4) + \frac{4(4 + 1)}{2} - 19 = 11$$

4. The smaller of U_1 and U_2 is the U value, i.e. 9.

As mentioned before, there are two steps in a Mann-Whitney U test.

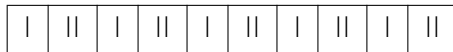
1. identify the U value, and
2. determine if the null hypothesis should be rejected.

Mann-Whitney U test

Our null hypothesis is that the medians of the two samples are equal.

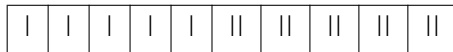
If the medians of the two samples are equal, the two samples will be evenly distributed across the rank of fitness. Therefore U_1 and U_2 will be similar and at a maximum value.

For example,



If the medians of the two samples differ a lot, the two samples will be separated clearly on the rank of fitness. U_1 and U_2 will be very different and the U value (the smaller value) will be small.

For example,



Thus, if the U value is small, we will reject the null hypothesis of the medians of the two samples being equal.

The question is, how small is 'small'?

This is the critical value that we will obtain from the Mann-Whitney U table, below which we will reject the null hypothesis.

Therefore, to determine if the null hypothesis should be rejected, we need to

1. identify the critical value, and
2. if the U value is less than the critical value, reject the null hypothesis, else, we do not reject the null hypothesis.

The critical values of the Mann-Whitney U for small values of n_1 and n_2 at significance level of 0.05 are given in the following table:

n_1	n_2							
	3	4	5	6	7	8	9	10
3	0	0	1	2	2	3	4	4
4	0	1	2	3	4	5	6	7
5	1	2	4	5	6	8	9	11
6	2	3	5	7	8	10	12	14
7	2	4	6	8	11	13	15	17
8	3	5	8	10	13	15	18	20
9	4	6	9	12	15	18	21	24
10	4	7	11	14	17	20	24	27

For $n_1 = 5$ and $n_2 = 4$, the critical value is 2.

As $U = 9 > 2$, we cannot reject the null hypothesis, and therefore we cannot say that one configuration produces fitness values that are significantly higher than the other.

Let say we test another configuration, which we named as C. We make five runs using configuration C and we obtain the fitness values of 0.060, 0.045, 0.043, 0.038, and 0.034. Compare with the fitness values of the four baseline cases (B), 0.102, 0.069, 0.055, and 0.049, what is the U value? Is the new configuration C statistically better than the baseline configuration B at 0.05 level of significance?

Notes:

- ▶ The previous table is only a partial table of critical values.
- ▶ When n_1 or n_2 is more than 20, the sampling distribution of U approaches the normal distribution.

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Consider a diagnostic or classification system with one output, i.e. the system will classify the input patterns to one of two classes.

The results can be illustrated as a contingency matrix.

		System Diagnosis	
		Positive	Negative
Gold Standard Diagnosis	Positive	true positive (TP)	false negative (FN)
	Negative	false positive (FP)	true negative (TN)

		System Diagnosis	
		Positive	Negative
Gold Standard Diagnosis	Positive	true positive (TP)	false negative (FN)
	Negative	false positive (FP)	true negative (TN)

Ratios:

- ▶ accuracy

$$\frac{TP + TN}{TP + TN + FP + FN}$$

- ▶ true positive ratio/sensitivity/recall

$$\frac{TP}{(TP + FN)}$$

- ▶ precision

$$\frac{TP}{(TP + FP)}$$

- ▶ false positive ratio/false alarm rate

$$\frac{FP}{(FP + TN)}$$

- ▶ true negative ratio/specificity

$$\frac{TN}{(FP + TN)}$$

- ▶ false positive ratio = 1 - specificity

Recall

- ▶ is the number of positive diagnoses correctly made by the system divided by the total number of positive diagnoses made by the gold standard, i.e. the true positive ratio.
- ▶ is also known as the probability of detection.
- ▶ provides an indication of the relative number of false negatives.

Precision (a.k.a. positive predictive value)

- ▶ is the number of positive diagnoses correctly made by the system divided by the total number of positive diagnoses made by the system, i.e. $\frac{TP}{TP+FP}$
- ▶ provides an indication of the magnitude of false positives.

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Confusion matrices

Confusion matrix is normally used when a system has multiple output classes.

Consider a system with 3 output classes.

		System outputs		
		A	B	C
Gold standard	A			
	B			
	C			

To construct a confusion matrix, we start by populating the table with the counts of output diagnoses/classes in each cell.

For example, we consider a medical diagnostic system with three diagnoses A, B and C. A neural network is trained.

Then we use 50 test cases of each diagnosis to obtain the following **confusion matrix**.

		System outputs		
		A	B	C
Gold standard	A	40	8	2
	B	6	42	2
	C	1	1	48

Next we divide each number by the total number of instances in its row to obtain the **class confusion matrix**.

		System outputs		
		A	B	C
Gold standard	A	0.80	0.16	0.04
	B	0.12	0.84	0.04
	C	0.02	0.02	0.96

Each row should add up to 1. This is the probability to be diagnosed as a certain class given that the case belongs to a specific diagnose. For example, the numbers in the first row are the probabilities of the case being diagnosed by the system to be A, B, or C given that the case is a case of A.

The diagonal values represent the recall of each class.

Then we multiply each element by the prior probability of the class of its row to obtain the **final confusion matrix**.

The prior probabilities are the probabilities of the occurrence of the case in actual situation.

For this case, say we have the prior probabilities as 0.60, 0.30, and 0.10 respectively.

		System outputs		
		A	B	C
Gold standard	A	0.480	0.096	0.024
	B	0.036	0.252	0.012
	C	0.002	0.002	0.096

Each element is now the probability of that outcome. The sum of all elements is now equal to 1.0.

The **accuracy** of the system is calculated by calculating the sum of the main diagonal values.

		System outputs		
		A	B	C
Gold standard	A	0.480	0.096	0.024
	B	0.036	0.252	0.012
	C	0.002	0.002	0.096

$$\text{accuracy} = 0.480 + 0.252 + 0.096 = 0.828$$

i.e. 82.8% of the diagnoses the system makes is correct.

Time to inject some reality. In real life, there are consequences to wrong diagnoses.

In medical diagnoses, the consequences are the treatment cost, human lives, legal cost, etc.

Therefore, we need to construct the cost matrix to investigate the cost of using the system.

Suppose that, the medical diagnosis represented by A is a condition that can be cured by an over-the-counter drug available at any drug store that costs \$10.

The medical diagnosis B is a more serious medical problem that requires aggressive treatment with a prescription antibiotic that costs \$100.

Condition C is a very serious condition that requires hospitalisation and surgery, with a total cost of \$5000. If condition C is not diagnosed right now, the patient has only a 20-percent chance of survival. Even though it is impossible to assign a dollar value to a human life, let's say that the average insurance policy of these patients is \$100,000. That means misdiagnosed condition C will result in an average cost of \$80,000 (80% chance of death $\times$ \$100,000).

It is often assumed that correct classification/diagnosis imposes zero cost, but it is clear that in our case, the cost for a correct diagnosis is not zero.

Now we are ready to identify the cost for each misdiagnosis and correct diagnosis based on the previous slide.

Cost matrix

		System outputs		
		A	B	C
Gold standard	A	\$10	\$110	\$5,010
	B	\$110	\$100	\$5,100
	C	\$80,010	\$80,100	\$5,000

Then we multiply this cost matrix with the final confusion matrix element by element to obtain the **final cost matrix**.

Final cost matrix

		System outputs		
		A	B	C
Gold standard	A	4.8	10.56	120.24
	B	3.96	25.2	61.2
	C	160.02	160.20	480.00

The average estimated cost of each application of the diagnostic system is given by the sum of all elements in the final cost matrix, which is \$1,026.18.

The cost calculations from confusion matrices can be used to fine-tune a computational intelligence system.

For example a neural network-based diagnostic system is first trained. Then the threshold levels of the output neurons can be adjusted by calculating the costs associated to choose the combination that produces the lowest system cost.

Even if cost information is not available, the system can still be tuned to minimise the off-diagonal numbers, of which the sum represents the percent incorrect.

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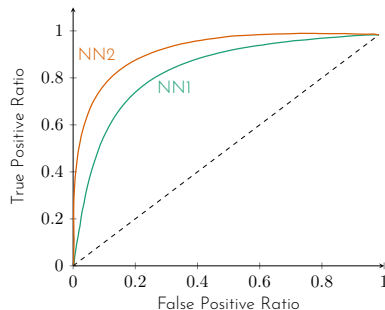
Chi-Square Test

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- A set of small navigation icons typically found in Beamer presentations, including symbols for back, forward, search, and other slide controls.

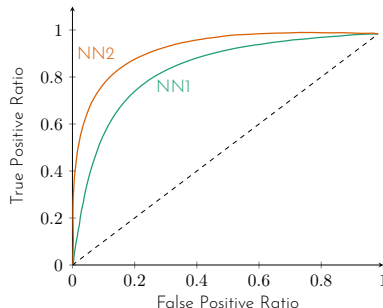
- ▶ An ROC curve corresponds to a system configuration.
- ▶ An ROC curve is a plot of the true positive ratio (sensitivity) versus the false positive ratio ($1 - \text{specificity}$).
- ▶ It is typical to plot the points for different values of output threshold, and connect them with line segments or a smooth curve to form the ROC curve.

Hypothetically the ROC curves for two neural networks are generated as follows:



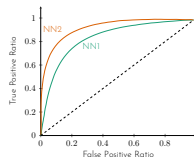
The questions of concern are:

- ▶ How do we interpret the ROC curves?
- ▶ Which is the neural network with better performance?



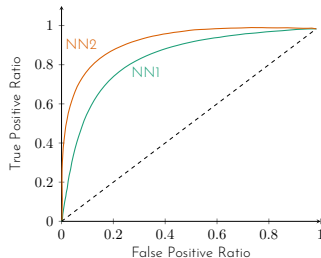
An ROC curve has two attributes:

- ▶ It always lies above the major diagonal.
- ▶ It is always monotonically increasing in value from left to right.



- ▶ The area under the ROC curve is used as a measure of the system performance.
 $0.5 < \text{area under ROC curve} < 1.0$
- ▶ 1.0 is the area corresponds to perfect performance.
- ▶ The larger the area is, the better the performance is.
- ▶ Calculation of the area under an ROC curve can be done by
 - generate a smooth curve through the points and calculate the area under it, or
 - join the points with straight lines and calculate the area using trapezoidal rule.

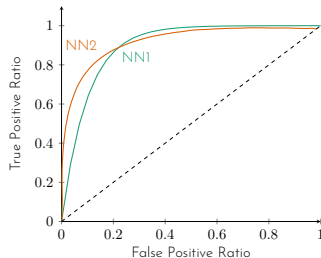
ROC curves



The shape of the curve indicates the sensitivity of the system performance to the threshold value.

Notes:

- ▶ Two ROC curves with same area may intersect. In such case, one performs better with respect to false positives, the other performs better with respect to false negatives.



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Chi-square (χ^2) test is used to evaluate the performance of the system where

- ▶ the 'gold standard' is unknown, i.e. we do not know the expected output,
- ▶ but **the expected frequency distribution of the categories is known**, i.e. we have the expected frequency of occurrence of each class or category.

To use chi-square test, the categories must be **mutually exclusive** and **collectively exhaustive**. The data is assumed to be normally distributed.

To perform chi-square test, we

1. determine the chi-square value χ^2 ,
2. determine the degree of freedom df ,
3. obtain critical values from the chi-square distribution table, and
4. determine if the null hypothesis (the expected frequency distribution and the observed frequency distribution has no significant difference) should be rejected.

To determine the chi-square value χ^2

For a system with n categories, E_i is the expected frequency of occurrence for category i and O_i is the observed frequency of occurrence for category i .

$$\chi^2 = \sum_{i=1}^n \frac{(O_i - E_i)^2}{E_i}$$

To determine the chi-square value χ^2

$$\chi^2 = \sum_{i=1}^n \frac{(O_i - E_i)^2}{E_i}$$

For example, in a test set of 50 patterns with 4 categories, the expected frequency distribution of the classification is 5, 10, 15, and 20.

The first system configuration produces results with frequency distribution of 6, 10, 14, and 20.

$$\chi_1^2 = \frac{(6 - 5)^2}{5} + \frac{(10 - 10)^2}{10} + \frac{(14 - 15)^2}{15} + \frac{(20 - 20)^2}{20} = 0.267$$

The second system configuration produces results with frequency distribution of 2, 15, 8, and 25.

$$\chi_2^2 = \frac{(2 - 5)^2}{5} + \frac{(15 - 10)^2}{10} + \frac{(8 - 15)^2}{15} + \frac{(25 - 20)^2}{20} = 8.82$$

To determine the degree of freedom df

For a system with n categories, the degree of freedom df is $n - 1$.

In our example, there are 4 categories, and therefore

$$df = 4 - 1 = 3$$

To obtain critical values from the chi-square distribution table

df	Area to the right									
	0.995	0.990	0.975	0.950	0.900	0.100	0.050	0.025	0.010	0.005
1	0.000	0.000	0.001	0.004	0.016	2.706	3.841	5.024	6.635	7.879
2	0.010	0.020	0.051	0.103	0.211	4.605	5.991	7.378	9.210	10.597
3	0.072	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.345	12.838
4	0.207	0.297	0.484	0.711	1.064	7.779	9.488	11.143	13.277	14.860
5	0.412	0.554	0.831	1.145	1.610	9.236	11.070	12.833	15.086	16.750
6	0.676	0.872	1.237	1.635	2.204	10.645	12.592	14.449	16.812	18.548
7	0.989	1.239	1.690	2.167	2.833	12.017	14.067	16.013	18.475	20.278
8	1.344	1.646	2.180	2.733	3.490	13.362	15.507	17.535	20.090	21.955
9	1.735	2.088	2.700	3.325	4.168	14.684	16.919	19.023	21.666	23.589
10	2.156	2.558	3.247	3.940	4.865	15.987	18.307	20.483	23.209	25.188
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮

For degree of freedom $df = 3$, at 0.05 level of significance, our critical value is 7.815.

To determine if the null hypothesis should be rejected

Our null hypothesis is “the expected frequency distribution and the observed frequency distribution has no significant difference”.

$$\chi^2 = \sum_{i=1}^n \frac{(O_i - E_i)^2}{E_i}$$

If there is no difference between the expected and the observed frequency distribution, we will have $\chi^2 = 0$.

If χ^2 is sufficiently big, it means that the expected and observed frequency distribution are different.

The boundary or threshold value for χ^2 for us to conclude that the difference between the two distributions is significant is the critical value.

Therefore,

If the χ^2 value is greater than or equal to the critical value,

we **reject** the null hypothesis,

whereas if the χ^2 value is less than the critical value,

we **do not reject** the null hypothesis.

To determine if the null hypothesis should be rejected

In our example,

$\chi_1^2 = 0.267$, $\chi_2^2 = 8.82$, and the critical value is 7.815.

$$\chi_1^2 = 0.267 < 7.815,$$

the null hypothesis is not rejected at 0.05 significance level for first configuration.

$$\chi_2^2 = 8.82 \geq 7.815,$$

the null hypothesis is rejected at 0.05 significance level for second configuration.

$$\chi_1^2 = 0.267 < 7.815,$$

the null hypothesis is not rejected at 0.05 significance level for first configuration.

This means that the difference between the expected frequency distribution and the observed frequency distribution in the first configuration is not significant.

$$\chi^2_2 = 8.82 \geq 7.815,$$

the null hypothesis is rejected at 0.05 significance level for second configuration.

This means that the expected frequency distribution and the observed frequency distribution in the second configuration are significantly different.

Therefore, we would say that the first system configuration is better than the second one.

Or at the very least, the first system configuration is acceptable while the second one is not.

Notes:

- ▶ Chi-square test evaluates the performance of the entire system at once, as oppose to ROC curve, which is designed to analyse one output neuron (one category) at a time.
- ▶ It is necessary to determine the categories that are possible. In some cases, not all categories are expected. The categories with zero expected frequency have to be handled with care, whether to remove them, or to include them and perform division by zero.

Notes:

- ▶ In general, chi-square test should only be used when $\leq 20\%$ of the expected counts are < 5 , and all individual counts are at least 1.
- ▶ It should be obvious that when designing or tuning a computational intelligence system, the goal is to minimise the chi-square value, i.e. the observed frequency distribution should be as similar to the expected frequency distribution as possible.

A classification system categorises objects into 3 categories, A, B, and C. A test set consists of 20 items for each class is used to perform testing on two classification algorithms. The result of algorithm I shows 30 A, 15 B, and 15 C whereas algorithm II shows 23 A, 18 B, and 19 C. Use χ^2 test to evaluate the two algorithms.

Introduction

Gold Standard

Percent Correct

Errors

Average Sum-Squared Error

Absolute Error

Normalised Error

Mann-Whitney U Test

Matrices

Contingency Matrices

Confusion Matrices

Receiver Operating Characteristic (ROC) Curves

Chi-Square Test



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Thank you

CSC3034 Current Course Update

10 Performance Metrics - handout

Synchronized from the current CSC3034 course site on 14 August 2026.

- Current labs connect core CI algorithms with physical-AI simulation examples.
- Isaac Sim examples use its bundled Python 3.12 environment and current launchers.
- Every simulator-dependent example includes a CPU/Matplotlib learning fallback.

Current materials author: Dr Tang Tiong Yew

See the synchronized lab notes and runnable examples for full instructions.